Inverse of High Dimension Matrix

Recall

$$\underline{W} = (X^T X + \lambda I)^{-1} X^T Y$$

if Feature vectors are of dimension d ⇒ (X^T X) is of size d+1 x d+1

The standard matrix inversion algorithm has a time complexity of O(n^3), which means that the computation time increases rapidly with the size of the matrix (increasing cubically with the number of components).

For Example: if Features vector is of size 1000 then Inverse calculation is or order 1,000,000,000 which is impractical

Solution:

Instead of using Least Squares with Matrix inversion,
use Gradient Descent iterative solution